

LECTURE NOTES MATH254

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1. LECTURE 2

In this class we introduce rings and fields.

1.1. Rings. In the last course we studied *groups* which are sets with one binary operation (satisfying the axioms of a group). In this course the focus will be on *rings* and *fields*, which are sets with two binary operations. We begin by giving the definition:

Definition 1.1 (Definition of a ring). *A ring $(R, +, \cdot)$ is a set R with two binary operations $+$ and $\cdot$ satisfying: $(R, +)$ is an abelian group, $\cdot$ is associative:*

$$(1.1) \quad (a \cdot b) \cdot c = a \cdot (b \cdot c), \quad a, b, c \in R,$$

multiplication distributes over addition:

$$(1.2) \quad a \cdot (b + c) = a \cdot b + a \cdot c, \quad (b + c) \cdot a = b \cdot a + c \cdot a, \quad a, b, c \in R.$$

Remark 1. The operation $+$ is called *addition*, and the operation $\cdot$ is called *multiplication*. We will denote the identity element for $+$ by 0 , so $a + 0 = 0 + a = a$ for all $a \in R$. Since $(R, +)$ is an abelian group, every element $a \in R$ has an inverse with respect to $+$, we will denote this element by $-a$. It is also common to not write out $\cdot$ in calculations (to save space), so we will often write $a \cdot b = ab$.

Example 1.1. The following sets are examples of rings with the usual addition and multiplication: $\mathbb{Z}$, $\mathbb{Q}$, $\mathbb{R}$, $\mathbb{C}$.

Example 1.2. For any integer $n \geq 1$ the set $n\mathbb{Z}$ is also an example of a ring with the usual operations of addition and multiplication. We simply need to note that if $nx, ny \in n\mathbb{Z}$ then $nx \cdot ny = n(nxy) \in n\mathbb{Z}$.

Example 1.3. Sets of functions tends to be rings. As an example, the set of C^∞ -functions $f : \mathbb{R} \rightarrow \mathbb{R}$, $C^\infty(\mathbb{R})$, is an example of a ring. Another set of functions that form a ring is the C^∞ -functions that vanish at infinity. Let $f \in C^\infty(\mathbb{R})$, we say that f vanish at infinity if:

$$(1.3) \quad \lim_{|t| \rightarrow \infty} f(t) = 0.$$

We denote the set of all smooth functions that vanish at infinity by $C_0^\infty(\mathbb{R})$, this is also an example of a ring (why? Show that sums and products of functions that vanish at infinity, vanish at infinity!).

Example 1.4. In the previous examples the multiplication was always commutative. An example of a ring where the multiplication is not commutative is the set of all $n \times n$ -matrices with real entries (and $n \geq 2$). Let $M_{n \times n}(\mathbb{R})$ be the set of all $n \times n$ -matrices with real entries, this forms a ring with addition of matrices and products of matrices as binary operations.

These are some of the basic important properties of rings:

Theorem 1.1. *If R is a ring with additive identity 0 , and $a, b \in R$ then we have:*

- (i) $0a = a0 = 0$,
- (ii) $a(-b) = (-a)b = -(ab)$,
- (iii) $(-a)(-b) = ab$.

Proof. Note that $0 + 0 = 0$, since 0 is an identity with respect to $+$. Using that products distributes over sums it follows that for any $a \in R$ we have:

$$(1.4) \quad 0a = (0 + 0)a = 0a + 0a$$

and since $0 = 0a + (-0a)$ we obtain:

$$(1.5) \quad 0 = 0a + (-0a) = 0a + 0a + (-0a) = 0a$$

or $0a = 0$. That $a0 = 0$ is proved in the same way (prove this!). To prove the second formula, we can use (i) and the fact that $\cdot$ distributes over $+$:

$$(1.6) \quad 0 = 0b = (a + (-a))b = ab + (-a)b = (-a)b + ab$$

so adding $-(ab)$ on both sides we cancel ab on the right and obtain $-(ab) = (-a)b$. That $a(-b) = -(ab)$ is proved completely analogously (prove this!). Finally we can use (ii) to show (iii):

$$(1.7) \quad (-a)(-b) = -(a(-b)) = -(-(ab)) = ab$$

where the last equality use that for any $c \in R$ we have $-(-c) = c$ since the inverse of the inverse of an element (in any group) is equal to the original element. ■

In any ring R , for any element $a \in R$, and for any integer $n \in \mathbb{Z}$, $n > 0$, we define $na = a + a + \dots + a$ where we are summing a n times. Similarly, we define for $-n$, when $n > 0$, $(-n)a = -(na)$, and we define $0a = 0$. This shorthand is often useful when we do calculations, however **be careful: the integer n may not be an element of R , so na is not the product in R .**

We end this subsection by two more definitions about rings that will be useful when defining fields.

Definition 1.2. A ring R is commutative if the binary operation $\cdot$ is commutative. That is, R is a commutative ring if for all $a, b \in R$ we have $ab = ba$.

Example 1.5. All examples of rings we have had are commutative, except for the ring of $n \times n$ -matrices (with $n \geq 2$). That is, $\mathbb{Z}, \mathbb{Q}, \mathbb{R}, \mathbb{C}, C^\infty(\mathbb{R}), C_0^\infty(\mathbb{R})$ are all commutative rings, while $M_{n \times n}(\mathbb{R})$ (with $n \geq 2$) is **not** a commutative ring.

Definition 1.3. A ring R with a multiplicative identity is a ring with unity. That is, R is a ring with unity if there is $1 \in R$ such that $1a = a1 = a$ for all $a \in R$. We call the element $1 \in R$ the unity of R .

Example 1.6. The ring $\mathbb{Z}, \mathbb{Q}, \mathbb{R}, \mathbb{C}, C^\infty(\mathbb{R}), M_{n \times n}(\mathbb{R})$ are all rings with unity where the multiplicative identity is 1 in the first 4 examples, the multiplicative identity is the constant function 1 in $C^\infty(\mathbb{R})$, and the multiplicative identity in $M_{n \times n}(\mathbb{R})$ is the identity matrix. The ring $n\mathbb{Z}$, $n \geq 2$, is not a ring with unity since $1 \notin n\mathbb{Z}$. The ring $C_0^\infty(\mathbb{R})$ is **not** a ring with unity since the constant function 1 does not vanish at infinity.

1.2. Fields. In a ring we can add and multiply elements, but we can not divide. One way to introduce the notion of division is through the notion of a *field*. We will first introduce the notion of a unit in a ring R .

Definition 1.4. Let R be a ring with unity and $0 \neq 1$. We say that $u \in R$ is a unit if there is u' such that $uu' = u'u = 1$. That is, $u \in R$ is a unit if u has a multiplicative inverse, in this case we will write u^{-1} for the inverse (or sometimes $1/u$).

Example 1.7. The ring $\mathbb{Z}$ has precisely two units: 1 and -1 .

Example 1.8. In $\mathbb{Q}, \mathbb{R}, \mathbb{C}$ every non-zero element is a unit: the inverse of x is simply $1/x$. An element in $A \in M_{n \times n}(\mathbb{R})$ is a unit if $A \in \text{GL}(n, \mathbb{R})$ and in this case the inverse is the inverse matrix A^{-1} .

Example 1.9. An element $f \in C^\infty(\mathbb{R})$ is a unit if and only if $f(x) \neq 0$ for all $x \in \mathbb{R}$, in this case the multiplicative inverse of f is given by $g(x) = 1/f(x)$. No element of $f \in C_0^\infty(\mathbb{R})$ is a unit (this would make no sense since $C_0^\infty(\mathbb{R})$ does not have a unity).

Definition 1.5. We say that a ring F is a field if: F is a commutative ring with unity such that $0 \neq 1$, and such that every non-zero element of F is a unit (that is, for every $a \in F \setminus \{0\}$ we find $b \in F$ such that $ab = ba = 1$).

As we have seen (Example 1.8) every non-zero element in $\mathbb{Q}, \mathbb{R}$, and $\mathbb{C}$ is a unit, so these are all examples of fields. A slightly more complicated example of a field is the following:

Example 1.10. Let p be a prime and $\mathbb{Z}_p = \mathbb{Z}/p\mathbb{Z}$. Recall that we can define multiplication and addition in $\mathbb{Z}_p$ by: $[x]_p \cdot [y]_p = [xy]_p$ and $[x]_p + [y]_p = [x + y]_p$. We showed in the homework of the previous course that every non-zero element in $\mathbb{Z}_p$ has a multiplicative inverse, so every non-zero element of $\mathbb{Z}_p$ is a unit. That is, $\mathbb{Z}_p$ is an example of a field (check the other conditions as well!).

Remark 2. In the remainder, we will simply write xy and $x + y$ for multiplication and addition of elements in $\mathbb{Z}_p$.

Example 1.11. The ring $\mathbb{Z}$ is not a field since only $1, -1$ are units, so, for example, the element $2 \in \mathbb{Z}$ is non-zero and has no multiplicative inverse. Indeed, the multiplicative inverse of 2 is $1/2$ but $1/2 \notin \mathbb{Z}$.

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